

Kaun Lab Protocol

Protocol Name: Two-Choice_graphing+_choice_behavior_code_

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Purpose of this protocol: This protocol will list instructions for running functions to use data collected from Two-Choice assay_TREX+_data aggregation code that outputs post-choice behavioral metrics like U-turns, distance to port and stops close to port, as well as make line graphs of behaviors over time.

NOTE: This code was designed to be ran in Google Colab in conjunction with data saved in Google Drive. The following instructions are written per box of code. It was meant to be used following the data aggregation code, with which you would have processed data of each timestamp folder you ran. Before doing anything, separate these processed timestamp folders by condition. I recommend making a copy of these folders and their contents and putting them (grouped by condition) in new folders named “EA” and “IAAc” for the purposes of running the following code. If you want to change the names of these folders, I recommend you make a copy of them and their contents to rename, as some of the code relies on identifying each folder by these names.

Box 1: Metrics per-side and per-quartile batch processing.....

This box will output csvs and PDF line graphs per-arena region (Side A vs. Side B) and broken into quartiles ordered by preference for each side for the following metrics:

Speed

Distance to port

Nearest border distance

Angular velocity

Acceleration

Angular acceleration

1. Make a new folder within each “EA” and “IAAc” folder that already contain timestamp folders of processed data per-animal generated in the “Batch-Processing” box in the data aggregation code (Step 6) and label them “metrics over time”.
2. In line 774 of box 1, paste the directory of the EA or IAAc folder in the first set of quotation marks. In the second set, paste the directory of the “metrics over time” folder corresponding with the EA or IAAc folder you put in the first set of quotations.
3. Set export_binned_csvs and plot_binned to “True” or “False” depending on whether you want them.

4. Set `quartile_basis` to either “Rolling_Preference_B” or “Rolling_Preference_A” depending on which side preference you want quartiles to be ordered by.
5. Set `transition_check_range` to the interval length (in frames) of the recordings you analyzed, then press the play button (circle with arrow at top left of box).

Box 2: per-odor preference index raw csvs and graphs.....

This box will generate line graphs and their respective csvs for preference index over time within each EA and IAAC generated folder from the data aggregation code.

1. In line 140, paste the directory of the EA or IAAC folder you are running in the first set of quotations, then paste the directory of where you’d like the generated csvs and graphs to save to in second set of quotations.
2. Box 2 also optionally outputs binned data and line graphs with customizable bin sizes, you can set `plot_binned` and `export_binned_csvs` to “True” or “False”. Figure size and bin size (in frames) and saved pdf DPI can be set as parameters in the variables following the pasted directories (lines 141 and 142). Press the play button when ready

Box 3: per-condition preference index graphs.....

This box uses the raw csvs generated in Box 2 to make line graphs depicting preference index over time per-condition.

1. Make copies of the `Raw_Ratio__Preference_Index_Over_Time.csvs`. They can be renamed to add additional descriptions of the conditions but do not change or remove the original label.
2. Make a new folder location to put these csvs in.
3. The csvs you will put in the new folder will depend on what condition was on what side, and what you would like to show on the graph. *For example*, if you want to show the preference index for a sucrose-paired odor, and sucrose-paired odor A always came from side A, and sucrose-paired odor B always came from side B, you would put the “`Raw_Ratio_A_Preference_Index_Over_Time`” from the EA folder and “`Raw_Ratio_B_Preference_Index_Over_Time`” from the IAAC folder in the new folder to make a graph plotting preference relative to the side with the sucrose-paired odor. This will work with any number of csvs.
4. In line 104, paste the directory of the new folder with the chosen csvs inserted. In line 105, paste the directory of the location you would like the graph to save to.

5. Lines 106 to 111 contain parameters to adjust the aesthetics of the graph, adjust to your preferences. You can also manually change aesthetics and labels in lines 53-58 and lines 67-93. Press the play button when ready.

Box 4. Bout duration per side csvs and graphs.....

This box gets bout duration (in seconds per 15 second bin) csvs and line graphs per side. *Note* that it *currently* assumes you want to merge the bout duration data relative to side A that's coming from the EA generated folder and relative to side B that's coming from the IAAC generated folder to make the EA_bout_length_ovr_time csv, and vice versa for the IAAC_bout_length_ovr_time csv. It does this to capture the bout durations on the sucrose paired odor when the animals in the EA generated folder received the sucrose paired odor on side A, and the animals in the IAAC generated folder received the sucrose paired odor on side B. Therefore, The "EA" line on the graph currently represents the bout duration on the sucrose-paired side and the "IAAC" line represents the no-suc paired side. You can change what side the code will take data from and the name of the folders it will look within for this data in lines 64-77.

1. In line 196, paste the directory to the EA generated folder inside the first set of quotations, then in 197 paste the directory to the IAAC generated folder inside the second set of quotations, then in 198 paste the directory to the location you'd like the output to save to in the third set of quotations.
2. The graph aesthetics and labels can be manually changed on lines 138-139 and 152-181. Press the play button when ready.

Box 5: transitions to each side csvs.....

This box will get the csvs of transitions to each side per bin (5, 10, and 15 seconds per bin) to be used to plot transitions over time. Runs one EA or IAAC generated folder at a time. Data

1. In line 100, paste the directory of the EA or IAAC generated folder. In line 101, paste the directory of the location you want the output csvs to be saved to. Press the play button when ready.

Box 6: total transitions per bin over time graph.....

This box takes the csvs per bin generated by box 5 and plots the average transitions made per bin over time. The 5, 10 and 15 second bin csvs will all work.

1. Place the two pairs of csvs (of the bin size you want) that were generated from box 5 into a new folder location. *Note* that the code is currently set up combine two pairs of csvs (for example A to B and B to A transitions from the EA generated folder) and will identify them by matching labels of “EA” or “IAAc” in their names. If you want to change the labels to match a certain combination of animals or transitions to a certain side that you’re interested in, ensure that you rename the two pairs to new labels that are matched by the pairs you’re interested in and change the “categories” in lines 65 and 66 for the code to identify the new labels and combine the appropriate csvs.
2. Aesthetics can be manually changed in lines 105-112 and 123-143.
3. Press the play button and follow instructions to set graph parameters that pop up in the lines just below the box.

Box 7: U-turns and pre-post-turn orientation csvs.....

This box outputs csvs for U-turns made on each side per-15 second bin (recommended minimum 15 second bins based on Two-Choice paper findings, and orientations in degrees before and after sharp turns per bin. It runs a single EA or IAAc generated folder at a time.

1. In line 198, paste the directory of the EA or IAAc generated folder to be ran,
2. In lines 199-205, paste the directories of the locations you would like each csv to be saved to and include the following csv titles for each respective line (199-205).

```
/orientation.csv
/orientation_after_turns_B.csv
/orientation-after_turns_A.csv
pre_turns_A.csv
/pre_turns_B.csv
away_midline_A.csv
/away_midline_B.csv
```

I haven’t tried naming these csvs anything else, it may be fine. “Away_midline_a” and “away_midline_b” are U-turns made on side A and B, respectively. Press the play button when ready

Box 8: Stops close to port and distance of each stop csvs.....

This box gets the number of close stops made per 15-second bin (bin size is recommended as 15 seconds but is customizable) (a close stop is currently defined as occurring within 0.9 cm of the given odor port) and how close each stop is to the port of the currently occupied side. Runs a single EA or IAAc generated folder at a time.

1. In line 128, paste the directory to the EA or IAAc generated folder inside the first set of quotations, in 129, paste the directory where you would like the output csvs to be saved.
2. Recording, close stop and stop definition parameters can be manually changed in lines 8-15. Press the play button when ready.

Box 9: regional distance travelled csvs.....

This box gets the distance travelled on each arena side per bin (bin size can be customized).

1. In line 135, paste the directory of the EA or IAAc generated folder within the first set of quotations, and in 136 paste the directory of the location you would like the output csvs to be saved.
2. Frames per bin and start and end frames can be manually adjusted in lines 10-15. *Note, don't worry* if you see the following:
No valid sharp turn data found.
No valid stop data found.

Turn and stop data were relegated to Boxes 7 and 8.

Box 10: regional distance travelled graph.....

This box uses the csvs generated from Box 9 to plot line graphs of regional distance travelled over time. *Note* that the code anticipates two pairs of csvs, 2 sets of distances travelled on side A and B in the EA generated folder, and the same two in the IAAc generated folder. It is currently set up to match pairs of csvs that have keywords "H2O" and "sucrose" in their names. These labels can be changed on lines 47 and 48. As in other boxes, change the names of pairs you want to combine from two groups to match each other, *for example*, if you want to compare the distance travelled on side A by the animals in both the EA and IAAc generated folders, to their distances travelled in side B, you would give those respective pairs a matching label in their name and change lines 47 and 48 to search for csvs with names containing that label.

1. Move (or make copies and move the copies) of the two pairs of csvs that were generated within the EA and IAAc folders in Box 9 to a new location, and paste the directory of that location in the set of quotations in line 119. Then paste the directory of the location where you'd like the graph to be saved in the set of quotations in line 120.
2. The aesthetics and parameters of the graph can be changed manually in lines 73-111. Press the play button when ready.